A Derivative-Free Bellman Equation for Dynamic Time-to-Collision: Two-Sided Certificates and Explicit ReLU Approximation

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***Abstract*— Time-to-collision (TTC) computed from the predicted evolution of a car-following model, rather than from frozen kinematics, defines a state feedback implicitly, through the first contact time of a prediction flow; evaluating it online requires a simulation at every control step. We study this feedback map as the value function V of a minimum-time problem for a Carathéodory flow with transversal first contact. Under hypotheses that allow merely bounded leader forcing, we prove that V is locally Lipschitz on the transversal collision domain, with a self-contained proof. We then characterize V by a stopped-step Bellman equation, which involves no derivatives and is evaluated by integrating the prediction for a fixed duration or until contact. Its solution is unique, and a two-sided a priori certificate bounds the error of any candidate by its residual along trajectories and its boundary error, with no differentiability required of the candidate; the fixed-point iteration recovers V exactly after finitely many sweeps from any initialization. We show, by contrast, that the optional-duration form of the equation used in earlier work is not identifiable: its fixed points are the dynamic-programming subsolutions, a convex family with V as its maximum. Finally, V admits a ReLU realization of explicit size whose only value-dependent weights are nodal TTC values, with controlled residual and robustness to nodal errors. The optimal velocity model is the running example.**

I. Introduction

Microscopic traffic models describe vehicle-to-vehicle interactions and underpin the analysis and control of driving behavior. Among them, the Optimal Velocity Model (OVM) reproduces stop-and-go waves and spontaneous congestion [1], [2], its properties are well understood [3]–[6], and it underlies many extensions [7]–[10]. One such extension incorporates a nonlinear Time-to-Collision (TTC) feedback into the OVM [11]. TTC is a surrogate safety measure that estimates the time until collision under simplified assumptions on the future inputs, such as constant velocities [12] or prescribed acceleration profiles [13]–[15]; it accounts jointly for spacing and closing speed [10], and drivers respond more strongly to it than to spacing or heading [16], [17]. The *dynamic-based* TTC of [11] instead predicts the collision time from the future evolution of a car-following model, which makes the feedback consistent with the model it is embedded in. Since the OVM admits no closed-form solution, each evaluation requires a numerical simulation, and the resulting delay can break the feedback loop; a real-time approximation of the dynamic-based TTC is therefore needed.

In [11], the TTC is identified with the value function $\boldsymbol{V}$ of a minimum-time Hamilton–Jacobi reachability problem [18], [19] and approximated by a physics-informed network [20] trained on the differential (HJB) residual, with a two-sided bound relating the training loss to the approximation error. The price is threefold: the residual involves derivatives of the candidate, so the certificate assumes a C^1 candidate class; the residual cannot vanish identically when the leader forcing is merely bounded; and the bound needs an additional term on the collision surface, whose analysis rests on a co-area and transversality argument. The same reference also introduced a derivative-free *weak* formulation, a Bellman operator taking an infimum over durations up to h; we show below that this form is not identifiable.

This paper studies the feedback map $\boldsymbol{V}$ through a different derivative-free equation, the *stopped-step* Bellman equation: advance the prediction for a fixed duration h, or until contact, whichever comes first, and add the elapsed time. The equation is evaluated exactly as the weak form is, needs no derivative of the candidate, and, unlike the weak form, has $\boldsymbol{V}$ as its unique solution with a two-sided residual certificate. Our contributions are:

- A self-contained regularity theory for the first-contact value function of a Carathéodory flow with transversal contact (Propositions 2 and 3, Appendix), allowing merely bounded time-dependent inputs and models defined only up to contact.
- The stopped-step equation: uniqueness of its solution, a two-sided a priori error bound in terms of the residual along trajectories, and exact recovery of $\boldsymbol{V}$ after finitely many sweeps from any initialization (Theorem 5 and Proposition 6). The certificate requires nothing of the candidate beyond boundedness.
- A characterization of the optional-duration (weak) form: its fixed points are exactly the dynamic-programming subsolutions, a convex family with maximum $\boldsymbol{V}$, so residual minimization for that form is ill-posed (Proposition 7).
- An optimization-free ReLU realization of $\boldsymbol{V}$ with explicit size, controlled stopped-step residual, and robustness to errors in the nodal values (Theorem 8).

The mechanisms are classical: the maximal-subsolution characterization of minimum-time functions [21], the linear-programming view of dynamic programming [22], [23], discrete dynamic programming for minimum-time problems [24], and ReLU interpolation [25], [26]. Unlike [**?**], which maximizes over HJB subsolutions for controlled problems with smooth data, there is no control here, the data are

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rough in time, and the object is a feedback map along a single prescribed flow, which is what makes the stopped-step equation identifiable and the certificate elementary.

II. Prediction Dynamics and Dynamic-Based TTC

A. Closed-loop model

The controlled system has state $\phi \in \mathbb{R}^p$ and dynamics

$$\dot{\phi} = \mathsf{F}\big(t, \phi, \tau(t,\phi)\big), \qquad \tau(t,\phi) \coloneqq \boldsymbol{V}\big(t, \pi(\phi)\big), \quad (1)$$

where τ is the dynamic-based TTC, $\boldsymbol{V}$ is the value function defined in (6) below from a prediction model, and $\pi : \mathbb{R}^p \to \mathbb{R}^d$ extracts from the actual state the initial condition of the prediction. Thus τ is a time-varying static state feedback, defined implicitly through the prediction model, as a model predictive controller is; this paper concerns the regularity of this implicit feedback map and its explicit, real-time computable approximation. We assume that (1) is well posed on $[0, \boldsymbol{T}_h]$ for the initial data of interest; this is established for the car-following instance below in [11]. Well-posedness is not used in this paper, whose results concern $\boldsymbol{V}$ alone: a lower bound on $\boldsymbol{V}$ is a lower bound on the first contact time *under the prediction model*, not a closed-loop statement.

Car-following instance. A platoon in which vehicle $n \in \{1, \dots, N\}$, with position x_n and velocity v_n, follows vehicle $n-1$; vehicle 0 is an exogenous leader with $\dot{v}_{\mathsf{L}} = \mathsf{a}_{\mathsf{L}}$, where

$$\mathsf{a}_{\mathsf{L}} \in L^\infty(0, \boldsymbol{T}_h), \quad 0 \le v_{\mathsf{L},\circ} + \int_0^t \mathsf{a}_{\mathsf{L}}\, dr \le v_{\max}, \quad t \in [0, \boldsymbol{T}_h], \quad (2)$$

$v_{\mathsf{L},\circ}$ being the initial leader velocity and $v_{\max}$ the maximal speed. The TTC-feedback OVM of [11] reads

$$\dot{x}_n = v_n, \qquad \dot{v}_n = \alpha\big\{\mathcal{V}(x_{n-1} - x_n) - v_n\big\} + \beta\, \frac{v_{n-1} - v_n}{\tau_n^2}, \quad (3)$$

with $\alpha, \beta > 0$, $\mathcal{V}(r) \coloneqq \tanh(r-2) + \tanh 2$ the smooth, bounded, increasing, 1-Lipschitz optimal velocity function [27], [28], and $\tau_n(t) = \boldsymbol{V}(t, \pi_n(\phi(t)))$ with $\pi_n(\phi) \coloneqq (x_{n-1} - x_n,\, v_{n-1} - v_n,\, v_{n-1})$. The prediction for the pair $(n-1, n)$ is conditional on the acceleration of vehicle $n-1$ over the horizon, prescribed as in (2); for a single leader–follower pair this is the leader input [8], [11].

B. Prediction dynamics

Let $U \subset \mathbb{R}^d$ be open (the state space of the prediction model), $g \in C^1(U)$ the *collision margin*, $\mathcal{Q} \coloneqq \{\boldsymbol{y} \in U : g(\boldsymbol{y}) \le 0\}$ the collision set, closed relative to U, and $P \coloneqq \{\boldsymbol{y} \in U : g(\boldsymbol{y}) \ge 0\}$ the region in which the model is physically well defined. The prediction model is

$$\dot{\boldsymbol{y}} = \mathsf{G}(t, \boldsymbol{y}), \qquad \mathsf{G} : [0, \boldsymbol{T}_h] \times P \to \mathbb{R}^d. \quad (4)$$

We write $z = (s, \boldsymbol{y})$ for a point of $[0, \boldsymbol{T}_h] \times U$, $m \coloneqq d + 1$, and $|\cdot|$ for Euclidean norms. A *solution in* P from $(s, \boldsymbol{y}) \in [0, \boldsymbol{T}_h] \times P$ is an absolutely continuous $x : [s, s+\sigma] \to P$, $0 \le \sigma \le \boldsymbol{T}_h - s$, with $x(s) = \boldsymbol{y}$ and $\dot{x} = \mathsf{G}(t, x)$ a.e.; it is *maximal* if it is not the restriction of a solution in P on a longer interval.

Assumption 1. (A1) G is measurable in t, and locally bounded and locally Lipschitz in $\boldsymbol{y}$ on P, up to the collision surface, with constants independent of t: for every compact $\boldsymbol{C} \subset P$ there are $M_{\boldsymbol{C}}, L_{\boldsymbol{C}}$ such that, for a.e. t and all $\boldsymbol{y}, \boldsymbol{y}' \in \boldsymbol{C}$, $|\mathsf{G}(t, \boldsymbol{y})| \le M_{\boldsymbol{C}}$ and $|\mathsf{G}(t, \boldsymbol{y}) - \mathsf{G}(t, \boldsymbol{y}')| \le L_{\boldsymbol{C}} |\boldsymbol{y} - \boldsymbol{y}'|$.

(A2) Forward completeness up to the boundary of P: from every $(s, \boldsymbol{y}) \in [0, \boldsymbol{T}_h] \times P$ there is a maximal solution in P, defined on a compact interval $[s, s + \sigma(s, \boldsymbol{y})]$, so that no solution escapes to infinity, or to ∂U, before reaching $\boldsymbol{T}_h$ or leaving P (if $s + \sigma < \boldsymbol{T}_h$ then $g(x(s+\sigma)) = 0$, since otherwise Carathéodory's existence theorem on a closed ball in $\{g > 0\}$ would continue the solution). For $g(\boldsymbol{y}) > 0$ the first contact time $\varsigma(s, \boldsymbol{y}) \le \sigma(s, \boldsymbol{y})$ is the first time the solution reaches $\mathcal{Q}$, or $\boldsymbol{T}_h - s$ if there is none, and the prediction is the solution on $[s, s + \varsigma(s, \boldsymbol{y})]$.

(A3) The normal velocity $\mathsf{n}(t, \boldsymbol{y}) \coloneqq \nabla g(\boldsymbol{y}) \cdot \mathsf{G}(t, \boldsymbol{y})$ is continuous on $[0, \boldsymbol{T}_h] \times P$.

By (A1) and Gronwall's inequality, two solutions in P from the same datum coincide on their common interval [29, Ch. 2, §1]. For $g(\boldsymbol{y}) > 0$ we denote by $\Phi(\kappa; s, \boldsymbol{y})$, $0 \le \kappa \le \varsigma(s, \boldsymbol{y})$, the state of the prediction at time $s + \kappa$. On this domain $\kappa \mapsto \Phi(\kappa; s, \boldsymbol{y})$ is Lipschitz, and the semigroup property $\Phi(\kappa_2; s + \kappa_1, \Phi(\kappa_1; s, \boldsymbol{y})) = \Phi(\kappa_1 + \kappa_2; s, \boldsymbol{y})$ holds whenever $\kappa_1 + \kappa_2 \le \varsigma(s, \boldsymbol{y})$. Nothing beyond contact is defined or used: the operator of Section III evaluates the flow only on $[0, \varsigma]$, and the proof of Proposition 3 uses a local extension of G that does not affect contact times. Condition (A3) says that the rate at which g changes along trajectories is continuous even though G is only measurable in t; it holds for second-order car-following models with Lipschitz acceleration laws and bounded inputs, where g is the gap and n the relative velocity. A uniform version of (A2), trajectories from a compact set of initial data staying in a compact subset of P, follows from (A1)–(A2) (Lemma 10 in the Appendix).

The OVM instance. For the car-following application, (4) is the OVM, i.e., (3) without the β-term, started from π_n of the current state. Denoting the predicted positions and velocities of leader and follower by $(\xi_{\mathsf{L}}, \zeta_{\mathsf{L}})$ and (ξ, ζ), the predicted gap $\Xi \coloneqq \xi_{\mathsf{L}} - \xi$ and relative velocity $\Lambda \coloneqq \zeta_{\mathsf{L}} - \zeta$, the state is $\boldsymbol{y} = (\Xi, \Lambda, \zeta_{\mathsf{L}}) \in \mathbb{R}^3$, $U = \mathbb{R}^3$, $g(\boldsymbol{y}) = \Xi$, $P = \{\Xi \ge 0\}$, and

$$\begin{aligned} &\dot{\Xi} = \Lambda, \qquad \dot{\zeta}_{\mathsf{L}} = \mathsf{a}_{\mathsf{L}}(t), \\ &\dot{\Lambda} = \mathsf{a}_{\mathsf{L}}(t) - \alpha\big\{\mathcal{V}(\Xi) - \zeta_{\mathsf{L}} + \Lambda\big\}. \end{aligned} \quad (5)$$

Assumption 1 holds: the forcing a_{L} enters additively and cancels in $\mathsf{G}(t, \boldsymbol{y}) - \mathsf{G}(t, \boldsymbol{y}')$, so the field is globally Lipschitz in $\boldsymbol{y}$ with a constant $1 + 2\alpha$ independent of t and, since $|\mathsf{G}(t, 0)|$ is bounded, of linear growth, which gives (A2) by Gronwall's inequality; and $\mathsf{n} = \Lambda$ is a state component, hence continuous. Here $m = 4$.

C. Dynamic-based TTC and the collision domain

The TTC is the value function of the first-contact problem

$$\boldsymbol{V}(s,\boldsymbol{y}) \coloneqq \min\Big\{\boldsymbol{T}_h - s,\ \inf\big\{\kappa \in [0,\varsigma(s,\boldsymbol{y})] : \Phi(\kappa;s,\boldsymbol{y}) \in \mathcal{Q}\big\}\Big\} \tag{6}$$

with $\inf \varnothing = +\infty$, i.e. $\boldsymbol{V}(s,\boldsymbol{y}) = \varsigma(s,\boldsymbol{y})$ for $g(\boldsymbol{y}) > 0$, and $\boldsymbol{V} \coloneqq 0$ on $\mathcal{Q}_T \coloneqq [0,\boldsymbol{T}_h] \times \mathcal{Q}$. Thus $0 \le \boldsymbol{V} \le \boldsymbol{T}_h - s$, with equality on the right exactly when no contact is predicted before $\boldsymbol{T}_h$. A tangential touch, $g = 0$ with $\mathsf{n} = 0$, counts as a contact: $\boldsymbol{V}$ is the time until the predicted margin first closes, whatever happens afterwards. Contacts are sought in the *transversal collision domain*

$$\mathbb{D} \coloneqq \big\{(s,\boldsymbol{y}) \in [0,\boldsymbol{T}_h) \times U : g(\boldsymbol{y}) > 0,\ \boldsymbol{V}(s,\boldsymbol{y}) < \boldsymbol{T}_h - s,\ \mathsf{n}\big(s + \boldsymbol{V}, \Phi(\boldsymbol{V};s,\boldsymbol{y})\big) < 0\big\}, \tag{7}$$

the states whose first contact with $\{g = 0\}$ is transversal. Writing $\Lambda(\kappa;s,\boldsymbol{y})$ for the Λ-component of $\Phi(\kappa;s,\boldsymbol{y})$, for the OVM the last condition reads $\Lambda(\boldsymbol{V};s,\boldsymbol{y}) < 0$: the predicted relative velocity is negative at impact. For $(s,\boldsymbol{y}) \in \mathbb{D}$ we call $(s+\boldsymbol{V}, \Phi(\boldsymbol{V};s,\boldsymbol{y})) \in \mathcal{Q}_T$ the *impact point*. We also use the stopped flow $\Phi^{\mathcal{Q}}(\kappa;s,\boldsymbol{y}) \coloneqq \Phi(\min\{\kappa, \boldsymbol{V}(s,\boldsymbol{y})\};s,\boldsymbol{y})$ for $g(\boldsymbol{y}) > 0$ and $\Phi^{\mathcal{Q}}(\kappa;s,\boldsymbol{y}) \coloneqq \boldsymbol{y}$ on $\mathcal{Q}$, which remains at the impact point once $\mathcal{Q}$ is reached, and $\Omega \coloneqq \mathbb{D} \cup \mathcal{Q}_T$.

Proposition 2 (Dynamic programming)**.** *Let $(s,\boldsymbol{y}) \in \mathbb{D}$ and $\nu \coloneqq \boldsymbol{V}(s,\boldsymbol{y})$. For every $\kappa \in [0,\nu)$, $(s+\kappa, \Phi(\kappa;s,\boldsymbol{y})) \in \mathbb{D}$ and $\boldsymbol{V}(s+\kappa, \Phi(\kappa;s,\boldsymbol{y})) = \nu - \kappa$.*

Proof. Let $z' \coloneqq (s+\kappa, \Phi(\kappa;s,\boldsymbol{y}))$. By the semigroup property, $\Phi(u;z') = \Phi(u+\kappa;s,\boldsymbol{y})$ for $0 \le u \le \nu - \kappa$. Since ν is the first contact time from $(s,\boldsymbol{y})$ and $\kappa < \nu$, we have $g(\Phi(\kappa;s,\boldsymbol{y})) > 0$, no contact from z' before $u = \nu - \kappa$, and a contact at $u = \nu - \kappa < \boldsymbol{T}_h - (s+\kappa)$. Hence $\boldsymbol{V}(z') = \nu - \kappa$, and the impact point of z' is that of $(s,\boldsymbol{y})$, where $\mathsf{n} < 0$. Therefore $z' \in \mathbb{D}$. □

Proposition 3 (Regularity)**.** *$\mathbb{D}$ is relatively open in $[0,\boldsymbol{T}_h) \times U$, and $\boldsymbol{V}$ is locally Lipschitz on $\mathbb{D}$, hence Lipschitz on every compact $\boldsymbol{K} \subset \mathbb{D}$, with a constant denoted $L_{\boldsymbol{V}}(\boldsymbol{K})$. Moreover $(\kappa,z) \mapsto \Phi(\kappa;z)$ is continuous on $\{(\kappa,z) : z \in \mathbb{D},\ 0 \le \kappa \le \boldsymbol{V}(z)\}$.*

Proof. Fix $z_0 = (s_0,\boldsymbol{y}_0) \in \mathbb{D}$, $\nu_0 \coloneqq \boldsymbol{V}(z_0) > 0$, the impact state $q \coloneqq \Phi(\nu_0;z_0)$, the impact time $t_* \coloneqq s_0 + \nu_0$, and $a \coloneqq -\mathsf{n}(t_*,q) > 0$.

Step 0: local extension. The pre-contact path $\Gamma \coloneqq \Phi([0,\nu_0];z_0)$ is a compact subset of U. Choose $\eta > 0$ such that the closed η-neighborhood Γ_η of Γ is a compact subset of U, and set $\boldsymbol{C} \coloneqq \Gamma_\eta \cap P$, a compact subset of P. Extend each component of $\mathsf{G}(t,\cdot)$ from $\boldsymbol{C}$ to $\mathbb{R}^d$ by the McShane formula with the constant $L_{\boldsymbol{C}}$ of (A1); on the null set of t where the inequalities of (A1) fail, set the extension to 0. The extended field $\tilde{\mathsf{G}}$ agrees with G on $\boldsymbol{C}$ for a.e. t, is measurable in t (infimum over a countable dense subset of $\boldsymbol{C}$), globally Lipschitz in $\boldsymbol{y}$ with constant $\tilde{L} \le \sqrt{d}\,L_{\boldsymbol{C}}$, and bounded by $M_{\boldsymbol{C}} + \tilde{L}|\boldsymbol{y} - q|$ near q; it has linear growth, so its flow $\tilde{\Phi}$ is defined on $[s,\boldsymbol{T}_h]$ for every datum and is jointly continuous. By uniqueness, $\tilde{\Phi}(\kappa;z) = \Phi(\kappa;z)$ whenever the latter stays in $\boldsymbol{C}$; in particular on $[0,\nu_0]$ for $z = z_0$. Set $F(z,\kappa) \coloneqq g(\tilde{\Phi}(\kappa;z))$ wherever $\tilde{\Phi}(\kappa;z) \in U$.

Step 1: Lipschitz dependence. Choose $\rho, \varepsilon_0 > 0$ with $s_0 + \rho < \boldsymbol{T}_h$, $2\rho \le \varepsilon_0$ and $\nu_0 + 2\varepsilon_0 \le \boldsymbol{T}_h - s_0 - \rho$, let $\mathcal{N} \coloneqq ([s_0 - \rho, s_0 + \rho] \cap [0,\boldsymbol{T}_h)) \times \bar{B}(\boldsymbol{y}_0,\rho)$ and $\bar{\kappa} \coloneqq \nu_0 + \varepsilon_0$, so that $\bar{\kappa} + \varepsilon_0 \le \boldsymbol{T}_h - s$ on $\mathcal{N}$. By joint continuity of $\tilde{\Phi}$, after shrinking ρ and ε_0 (keeping $2\rho \le \varepsilon_0$), the points $\tilde{\Phi}(\kappa;z)$, $z \in \mathcal{N}$, $0 \le \kappa \le \bar{\kappa} + \varepsilon_0$, lie in the interior of Γ_η, where g is Lipschitz with a constant L_g, and $|\tilde{\mathsf{G}}| \le \tilde{M}$ there. Then $|\tilde{\Phi}(\kappa;z) - \tilde{\Phi}(\kappa';z)| \le \tilde{M}|\kappa - \kappa'|$, Gronwall's inequality gives $|\tilde{\Phi}(\kappa;s,\boldsymbol{y}) - \tilde{\Phi}(\kappa;s,\boldsymbol{y}')| \le e^{\tilde{L}\bar{\kappa}}|\boldsymbol{y} - \boldsymbol{y}'|$ for $\kappa \le \bar{\kappa}$, and, for $s' = s + \delta$ with both $(s,\boldsymbol{y}),(s',\boldsymbol{y}) \in \mathcal{N}$, the bound $\bar{\kappa} + \varepsilon_0 \le \boldsymbol{T}_h - s'$ and $\delta \le 2\rho \le \varepsilon_0$ give $\kappa + \delta < \boldsymbol{T}_h - s$ and $\kappa + \delta \le \bar{\kappa} + \varepsilon_0$, so the semigroup identity $\tilde{\Phi}(\kappa + \delta;s,\boldsymbol{y}) = \tilde{\Phi}(\kappa;s',\tilde{\Phi}(\delta;s,\boldsymbol{y}))$ and Gronwall on $[0,\kappa]$ yield $|\tilde{\Phi}(\kappa;s',\boldsymbol{y}) - \tilde{\Phi}(\kappa;s,\boldsymbol{y})| \le (e^{\tilde{L}\bar{\kappa}} + 1)\tilde{M}\delta$. Since $\mathcal{N}$ is a product, F is Lipschitz on $\mathcal{N} \times [0,\bar{\kappa}]$ with a constant L_F.

Step 2: strict decrease near impact. For $x \in \mathbb{R}^d$ and a.e. t,

$$\nabla g(x)\cdot\tilde{\mathsf{G}}(t,x) - \mathsf{n}(t,q) = [\nabla g(x) - \nabla g(q)]\cdot\tilde{\mathsf{G}}(t,x) + \nabla g(q)\cdot[\tilde{\mathsf{G}}(t,x) - \mathsf{G}(t,q)],$$

using $\tilde{\mathsf{G}}(t,q) = \mathsf{G}(t,q)$ for a.e. t. By continuity of ∇g at q, the global Lipschitz bound of $\tilde{\mathsf{G}}$ and its local boundedness, there is $\rho_1 > 0$ with $\bar{B}(q,\rho_1) \subset U$ such that the right-hand side is at most $a/4$ in absolute value for a.e. t and all $|x - q| \le \rho_1$. By (A3) there is $\varepsilon_1 \in (0, \boldsymbol{T}_h - t_*]$ with $\mathsf{n}(t,q) \le -3a/4$ for $|t - t_*| \le \varepsilon_1$; shrinking ρ_1, ε_1, also $\mathsf{n} < 0$ on $[t_* - \varepsilon_1, t_* + \varepsilon_1] \times (\bar{B}(q,\rho_1) \cap P)$ by continuity of n. Hence

$$\nabla g(x)\cdot\tilde{\mathsf{G}}(t,x) \le -c \coloneqq -a/2 \quad \text{for a.e. } t \in [t_* - \varepsilon_1, t_* + \varepsilon_1] \text{ and all } |x - q| \le \rho_1. \tag{8}$$

Fix $\varepsilon \in (0, \min\{\varepsilon_0, \varepsilon_1/2, \nu_0, \rho_1/(2\tilde{M})\}]$, so that $\tilde{\Phi}(\kappa;z_0) \in \bar{B}(q,\rho_1/2)$ for $|\kappa - \nu_0| \le \varepsilon$ by Step 1, and shrink $\mathcal{N}$ to a compact relative neighborhood $\mathcal{N}'$ of z_0, of s-radius at most $\varepsilon_1/2$, such that $\tilde{\Phi}(\kappa;z) \in \bar{B}(q,\rho_1)$ for $z \in \mathcal{N}'$ and $\kappa \in [\nu_0 - \varepsilon, \nu_0 + \varepsilon]$ (joint continuity). For such $z = (s,\boldsymbol{y})$ and κ, $|s + \kappa - t_*| \le \varepsilon_1$, so (8) applies along the trajectory. Since $\kappa \mapsto g(\tilde{\Phi}(\kappa;z))$ is absolutely continuous with derivative $\nabla g(\tilde{\Phi})\cdot\tilde{\mathsf{G}}(s+\kappa,\tilde{\Phi})$ a.e., (8) integrates to

$$F(z,\kappa_2) - F(z,\kappa_1) \le -c(\kappa_2 - \kappa_1), \qquad \nu_0 - \varepsilon \le \kappa_1 \le \kappa_2 \le \nu_0 + \varepsilon, \tag{9}$$

for all $z \in \mathcal{N}'$.

Step 3: the first contact time. Since $F(z_0,\cdot) > 0$ on the compact interval $[0,\nu_0 - \varepsilon]$ and F is Lipschitz, after shrinking $\mathcal{N}'$ we have $F(z,\kappa) > 0$ for $z \in \mathcal{N}'$, $\kappa \in [0,\nu_0 - \varepsilon]$. By (9), $F(z_0,\nu_0 + \varepsilon) \le -c\varepsilon$, so after shrinking $\mathcal{N}'$ once more, $F(z,\nu_0 + \varepsilon) < 0$ for $z \in \mathcal{N}'$. Hence $F(z,\cdot)$ has exactly one zero $\kappa^*(z) \in (\nu_0 - \varepsilon, \nu_0 + \varepsilon)$, is positive before it, and $\kappa^*(z) < \bar{\kappa} < \boldsymbol{T}_h - s$. Before $\kappa^*(z)$ the extended trajectory lies in $P \cap \Gamma_\eta = \boldsymbol{C}$, where $\tilde{\mathsf{G}} = \mathsf{G}$, so by uniqueness it is the prediction from z, and $\boldsymbol{V}(z) = \varsigma(z) = \kappa^*(z)$; in particular $\Phi(\kappa;z) = \tilde{\Phi}(\kappa;z)$ for $z \in \mathcal{N}'$, $0 \le \kappa \le \boldsymbol{V}(z)$, which gives

the continuity claim. Moreover $g(\boldsymbol{y}) = F(z,0) > 0$, and the impact point of z lies in $\bar{B}(q,\rho_1)$ at a time in $[t_*-\varepsilon_1, t_*+\varepsilon_1]$, where $\mathsf{n} < 0$ by the choice of ρ_1, ε_1. Hence $\mathcal{N}' \subset \mathbb{D}$, which proves relative openness.

Step 4: Lipschitz bound. For $z, z' \in \mathcal{N}'$, (9) gives $|F(z', \kappa^*(z))| = |F(z', \kappa^*(z)) - F(z', \kappa^*(z'))| \geq c|\kappa^*(z) - \kappa^*(z')|$, while Step 1 gives $|F(z', \kappa^*(z))| = |F(z', \kappa^*(z)) - F(z, \kappa^*(z))| \leq L_F|z - z'|$. Hence $|\boldsymbol{V}(z) - \boldsymbol{V}(z')| \leq (L_F/c)|z - z'|$. A locally Lipschitz function is Lipschitz on compact subsets, by a standard covering argument. □

In particular $\boldsymbol{V}$ is continuous on $\mathbb{D}$; since $\mathbb{D}$ is relatively open, hence Borel, and $\mathcal{Q}_T$ is closed and disjoint from $\mathbb{D}$, $\boldsymbol{V}$ is Borel on Ω. The constants degenerate as $\boldsymbol{K}$ approaches $\partial\mathbb{D}$, where impacts become tangential and $\boldsymbol{V}$ may be discontinuous; the resulting discontinuity of the feedback in (1) concerns the closed loop, analyzed in [11].

III. The Stopped-Step Bellman Equation

Let $\mathscr{B}_0$ be the set of bounded functions $W : \Omega \to \mathbb{R}$ with $W = 0$ on $\mathcal{Q}_T$; functions are evaluated pointwise along trajectories, so we work with bounded functions rather than L^∞ classes, and $\boldsymbol{V} \in \mathscr{B}_0$ pointwise. Fix a step $h > 0$. For $z = (s, \boldsymbol{y}) \in \mathbb{D}$ let

$$\ell_h(z) \coloneqq \min\{h, \boldsymbol{V}(z)\}, \qquad z_\kappa \coloneqq \big(s+\kappa, \Phi^{\mathcal{Q}}(\kappa; z)\big), \tag{10}$$

and define, for a bounded $W : \Omega \to \mathbb{R}$, $S_h[W] \coloneqq 0$ on $\mathcal{Q}_T$ and

$$S_h[W](z) \coloneqq \ell_h(z) + W\big(z_{\ell_h(z)}\big), \qquad z \in \mathbb{D}. \tag{11}$$

Operationally, (11) integrates the prediction for duration h, or until contact if that occurs first, and adds the elapsed time; no knowledge of $\boldsymbol{V}$ is needed beyond event detection, and no derivative of W enters. The evaluation point $z_{\ell_h(z)}$ lies in Ω: in $\mathbb{D}$ if $h < \boldsymbol{V}(z)$ by Proposition 2, and at the impact point, in $\mathcal{Q}_T$, otherwise. Hence S_h maps bounded functions on Ω to bounded functions and $\mathscr{B}_0$ into itself; it is monotone, $W_1 \leq W_2 \Rightarrow S_h[W_1] \leq S_h[W_2]$, and non-expansive,

$$\big|S_h[W_1] - S_h[W_2]\big|(z) \leq |W_1 - W_2|\big(z_{\ell_h(z)}\big), \quad z \in \mathbb{D}. \tag{12}$$

Lemma 4 (Fixed point)**.** *$S_h[\boldsymbol{V}] = \boldsymbol{V}$ on Ω for every $h > 0$.*

Proof. On $\mathcal{Q}_T$ both sides vanish. For $z \in \mathbb{D}$ with $\nu \coloneqq \boldsymbol{V}(z)$: if $h < \nu$, Proposition 2 gives $\boldsymbol{V}(z_h) = \nu - h$, so $S_h[\boldsymbol{V}](z) = h + \nu - h = \nu$; if $h \geq \nu$, then $\ell_h(z) = \nu$, z_ν is the impact point, $\boldsymbol{V}(z_\nu) = 0$, and $S_h[\boldsymbol{V}](z) = \nu$. □

Theorem 5 (Uniqueness and two-sided certificate)**.** *Let $\boldsymbol{A} \subset \mathbb{D}$ contain the trajectories of its points up to impact, i.e. $(s+\kappa, \Phi(\kappa; s, \boldsymbol{y})) \in \boldsymbol{A}$ whenever $(s, \boldsymbol{y}) \in \boldsymbol{A}$ and $0 \leq \kappa < \boldsymbol{V}(s, \boldsymbol{y})$. Let $W : \Omega \to \mathbb{R}$ be bounded with $|W| \leq b$ on $\mathcal{Q}_T$ and*

$$\big|W - S_h[W]\big| \leq \delta \quad \text{on } \boldsymbol{A}. \tag{13}$$

Then, for all $z \in \boldsymbol{A}$,

$$|W(z) - \boldsymbol{V}(z)| \leq \big\lceil \boldsymbol{V}(z)/h \big\rceil \, \delta + b. \tag{14}$$

In particular, $\boldsymbol{V}$ is the unique fixed point of S_h in $\mathscr{B}_0$.

Proof. Fix $z \in \boldsymbol{A}$, $\nu \coloneqq \boldsymbol{V}(z)$, $N \coloneqq \lceil \nu/h \rceil$, and define $z_0 \coloneqq z$, $z_{j+1} \coloneqq (z_j)_{\ell_h(z_j)}$ for $j = 0, \ldots, N-1$. By Proposition 2, $z_j \in \boldsymbol{A}$ with $\boldsymbol{V}(z_j) = \nu - jh$ for $j < N$, so $\ell_h(z_j) = h$ for $j \leq N-2$, $\ell_h(z_{N-1}) = \nu - (N-1)h \in (0, h]$, and z_N is the impact point, in $\mathcal{Q}_T$. Set $e_j \coloneqq W(z_j) - \ell_h(z_j) - W(z_{j+1})$, so that $|e_j| \leq \delta$ by (13). Summing, $W(z) - \sum_{j<N} \ell_h(z_j) - W(z_N) = \sum_{j<N} e_j$, and $\sum_{j<N} \ell_h(z_j) = \nu$. Hence $|W(z) - \nu| \leq |W(z_N)| + N\delta \leq b + N\delta$. Uniqueness follows with $\boldsymbol{A} = \mathbb{D}$, $\delta = b = 0$. □

Theorem 5 is an a priori certificate: any candidate whose stopped-step residual is at most δ along every pre-contact trajectory of $\boldsymbol{A}$, and which is at most b on the collision set, is within $\lceil \boldsymbol{T}_h/h \rceil \delta + b$ of the TTC on $\boldsymbol{A}$. It requires nothing of W beyond boundedness, and nothing of $\boldsymbol{V}$ beyond Proposition 2; in particular no derivative of the candidate enters, in contrast with the HJB residual. The hypothesis on $\boldsymbol{A}$ cannot be dropped: the residual must be controlled along complete trajectories, not only on a sampled set; a one-sided version holds with one-sided residuals.

Proposition 6 (Exact iteration)**.** *For every $W_0 \in \mathscr{B}_0$ and $k \in \mathbb{N}$, $S_h^k[W_0] = \boldsymbol{V}$ on $\{z \in \mathbb{D} : \boldsymbol{V}(z) \leq kh\}$; in particular $S_h^{K_h}[W_0] = \boldsymbol{V}$ on Ω for $K_h \coloneqq \lceil \boldsymbol{T}_h/h \rceil$. For an arbitrary bounded $W_0 : \Omega \to \mathbb{R}$ the same holds with k replaced by $k+1$, since $S_h[W_0] \in \mathscr{B}_0$.*

Proof. With the notation of the previous proof, unrolling the iteration from z with $\nu \leq kh$ follows the chain $z_0, \ldots, z_N$, $N \leq k$, reaches the impact point $z_N \in \mathcal{Q}_T$ where every iterate is 0, and collects $\sum_{j<N} \ell_h(z_j) = \nu$; later applications of S_h on $\mathcal{Q}_T$ leave the value 0 unchanged. □

Exact collision information thus propagates backward from the target: after k sweeps the iterate is exact on the safety-critical set $\{\boldsymbol{V} \leq kh\}$ whatever the initialization. This is an identity for the continuum operator with exact flow and event evaluation; a numerical implementation with interpolation between nodes incurs the interpolation error at each sweep.

Proposition 7 (The optional-duration form is not identifiable)**.** *Let $\mathcal{T}_h[W](z) \coloneqq \inf_{0 \leq \kappa \leq \ell_h(z)} \{\kappa + W(z_\kappa)\}$ on $\mathbb{D}$ and $\mathcal{T}_h[W] \coloneqq 0$ on $\mathcal{Q}_T$, the derivative-free weak form of [11]. Then:*

- *(i) $\mathcal{T}_h[W] \leq W$ for every $W \in \mathscr{B}_0$, and $W = \mathcal{T}_h[W]$ if and only if $W(z) \leq \kappa + W(z_\kappa)$ for all $z \in \mathbb{D}$, $0 \leq \kappa \leq \ell_h(z)$; by the semigroup property, if and only if this holds for all $0 \leq \kappa \leq \boldsymbol{V}(z)$. The set $\mathfrak{S}$ of such W does not depend on h.*
- *(ii) $\mathfrak{S}$ is convex, closed under pointwise maxima and minima, and contains $\lambda\boldsymbol{V}$ for every $\lambda \in [0,1]$, in particular 0.*
- *(iii) $\boldsymbol{V} = \max \mathfrak{S}$: every $W \in \mathfrak{S}$ satisfies $W \leq \boldsymbol{V}$ on $\mathbb{D}$.*

Proof. (i) $\kappa = 0$ is admissible. For the second equivalence, split $\kappa \leq \boldsymbol{V}(z)$ into steps of length at most h along the trajectory and chain the inequalities using Proposition 2. (ii) The inequality is preserved by convex combinations, maxima and minima; for $W = \lambda\boldsymbol{V}$ it reads $\lambda\nu \leq \kappa + \lambda(\nu - \kappa)$,

i.e. $(1-\lambda)\kappa \geq 0$. (iii) Take $\kappa = \boldsymbol{V}(z)$: $W(z) \leq \boldsymbol{V}(z) + W(z_\nu) = \boldsymbol{V}(z)$. □

Thus the fixed points of the optional-duration operator are exactly the dynamic-programming subsolutions, a convex family of lower bounds on the TTC with $\boldsymbol{V}$ as its maximum. Minimizing the residual of $\mathcal{T}_h$ is therefore ill-posed: $W \equiv 0$ attains zero residual and satisfies the boundary condition. The one-dimensional example $\dot{y} = -1$, $\mathcal{Q} = \{y \leq 0\}$, $\boldsymbol{V}(s,y) = y$ for $0 < y < \boldsymbol{T}_h - s$, shows the contrast: for $W = \lambda \boldsymbol{V}$ and $y > h$, the true error is $|1-\lambda|y$, the optional-duration residual is $(\lambda-1)_+ h$, zero for all $\lambda \leq 1$, while the stopped-step residual is $|1-\lambda|h$, nonzero whenever $W \neq \boldsymbol{V}$. The deficiency is specific to the infimum: removing $\kappa = 0$ alone does not cure it for continuous W, since durations arbitrarily close to 0 remain.

IV. Explicit ReLU Realization

Let $\boldsymbol{D} \subset \mathbb{D}$ be compact and $\boldsymbol{V}_{\min}(\boldsymbol{D}) \coloneqq \min_{\boldsymbol{D}} \boldsymbol{V} > 0$. Fix $h \in (0, \boldsymbol{V}_{\min}(\boldsymbol{D}))$, so that $\ell_h = h$ on $\boldsymbol{D}$, and define the one-step forward tube and the enlarged set

$$\boldsymbol{D}^h \coloneqq \big\{(s+\kappa, \Phi(\kappa; s, \boldsymbol{y})) : (s, \boldsymbol{y}) \in \boldsymbol{D},\ \kappa \in [0,h]\big\},$$
$$\boldsymbol{D}' \coloneqq \boldsymbol{D} \cup \boldsymbol{D}^h.$$

$\boldsymbol{D}'$ is compact as a continuous image of $\boldsymbol{D} \times [0,h]$, and $\boldsymbol{D}' \subset \mathbb{D}$ by Proposition 2; for $z \in \boldsymbol{D}$, the operator S_h evaluates its argument at $z_h \in \boldsymbol{D}'$. Since $\boldsymbol{D}'$ is a compact subset of $[0, \boldsymbol{T}_h) \times U$ and $\mathbb{D}$ is relatively open, there is $r \in (0, \boldsymbol{T}_h - \max_{\boldsymbol{D}'} s)$, smaller than the distance from the $\boldsymbol{y}$-projection of $\boldsymbol{D}'$ to $\mathbb{R}^d \backslash U$, such that $\boldsymbol{K} \coloneqq \{z \in [0, \boldsymbol{T}_h) \times U : \operatorname{dist}_\infty(z, \boldsymbol{D}') \leq r\}$, with $\operatorname{dist}_\infty$ the max-norm distance, is a compact subset of $\mathbb{D}$. Let $L \coloneqq L_{\boldsymbol{V}}(\boldsymbol{K})$, let $\boldsymbol{B} \supset \boldsymbol{K}$ be a closed box with sides $\ell_1, \dots, \ell_m \leq \ell_{\boldsymbol{B}}$ whose s-projection lies in $[0, \boldsymbol{T}_h)$, and set $\Theta \coloneqq \max\{\boldsymbol{T}_h, L\ell_{\boldsymbol{B}}\}$.

Theorem 8 (Optimization-free ReLU realization)**.** *There is a constant C_m depending only on m such that, for every $\varepsilon \in (0, \Theta)$, there exist grid points $\{z_j\}_{j \in \mathcal{J}} \subset \boldsymbol{K}$ and ReLU networks $\{\tilde{\varphi}_j\}_{j \in \mathcal{J}}$, depending only on $(\boldsymbol{B}, \boldsymbol{D}', r, m, \varepsilon, \boldsymbol{T}_h, L)$ and not on the values of $\boldsymbol{V}$, such that the network*

$$\boldsymbol{V}_{\theta^*}(z) \coloneqq \sum_{j \in \mathcal{J}} c_j\, \tilde{\varphi}_j(z), \qquad c_j \coloneqq \boldsymbol{V}(z_j), \tag{15}$$

has depth at most $C_m(\ln(\Theta/\varepsilon) + 1)$ and at most

$$\mathcal{N}(\varepsilon, \boldsymbol{D}') \leq C_m \Big(\frac{\Theta}{\varepsilon} + \frac{\ell_{\boldsymbol{B}}}{r}\Big)^m \Big(\ln \frac{\Theta}{\varepsilon} + 1\Big) \tag{16}$$

weights and units in a sparse realization, and satisfies

$$\|\boldsymbol{V}_{\theta^*} - \boldsymbol{V}\|_{L^\infty(\boldsymbol{D}')} \leq \varepsilon, \tag{17}$$
$$\|\boldsymbol{V}_{\theta^*} - S_h[\boldsymbol{V}_{\theta^*}]\|_{L^\infty(\boldsymbol{D})} \leq 2\varepsilon. \tag{18}$$

If the nodal values are replaced by $\tilde{c}_j$ with $|\tilde{c}_j - \boldsymbol{V}(z_j)| \leq \eta$, then (17) *and* (18) *hold with ε replaced by $\varepsilon + 2^{m+1}\eta$.*

Proof. Step 1: interpolation. Fix $h_0 \in (0, r]$ and, on the ith axis of $\boldsymbol{B}$, take the uniform grid of mesh $h_i \coloneqq \ell_i / \lceil \ell_i / h_0 \rceil \leq h_0$, which divides the side ℓ_i; the number of nodes on that axis is $\lceil \ell_i/h_0 \rceil + 1 \leq \ell_i/h_0 + 2$. Let $\{z_j\}$ be the resulting tensor grid and $\varphi_j(z) \coloneqq \prod_{i=1}^m \max\{0, 1 - |z^i - z_j^i|/h_i\}$ the tensor-product hat functions: $\varphi_j \geq 0$, $\sum_j \varphi_j = 1$ on $\boldsymbol{B}$, and $\{\varphi_j \neq 0\} \subset \{|z - z_j|_\infty < h_0\}$, so at most 2^m of them are nonzero at any point. Keep in $\mathcal{J}$ only the nodes with $\varphi_j \not\equiv 0$ on $\boldsymbol{D}'$; since $h_0 \leq r$ and the s-projection of $\boldsymbol{B}$ lies in $[0, \boldsymbol{T}_h)$, these nodes lie in $\boldsymbol{K}$. For $z \in \boldsymbol{D}'$, since $|z - z_j| \leq \sqrt{m}\, h_0$ wherever $\varphi_j(z) \neq 0$, $|\boldsymbol{V}(z) - \sum_j \boldsymbol{V}(z_j)\varphi_j(z)| \leq \sum_j |\boldsymbol{V}(z) - \boldsymbol{V}(z_j)|\varphi_j(z) \leq L\sqrt{m}\, h_0$.

Step 2: ReLU realization. Each univariate hat is a one-layer ReLU network. By [25, Prop. 3 and proof of Thm. 1], the product of m factors in $[0,1]$ is approximated within $\varrho \in (0,1]$ by a ReLU network of depth and size $O(\ln(1/\varrho) + 1)$ that vanishes whenever one factor vanishes. This yields $\tilde{\varphi}_j$ with $|\tilde{\varphi}_j - \varphi_j| \leq \varrho$, $\tilde{\varphi}_j = 0$ wherever $\varphi_j = 0$, and $|\tilde{\varphi}_j| \leq 2$. As $0 \leq c_j \leq \boldsymbol{T}_h$ and at most 2^m terms are nonzero, $|\sum_j c_j(\tilde{\varphi}_j - \varphi_j)| \leq 2^m \boldsymbol{T}_h \varrho$. The choices $h_0 \coloneqq r$ if $L\sqrt{m}\, r \leq \varepsilon/2$ and $h_0 \coloneqq \varepsilon/(2L\sqrt{m})$ otherwise, and $\varrho \coloneqq \min\{1, \varepsilon/(2^{m+1}\boldsymbol{T}_h)\}$, give (17). The grid has at most $\prod_i (\ell_i/h_0 + 2) \leq (\ell_{\boldsymbol{B}}/r + 2\sqrt{m}\,\Theta/\varepsilon + 2)^m$ nodes, each costing $O(\ln(1/\varrho) + 1)$ units with $\ln(1/\varrho) \leq \ln(\Theta/\varepsilon) + (m+1)\ln 2$; since $\Theta/\varepsilon > 1$, this gives (16).

Step 3: residual. For $z \in \boldsymbol{D}$, Lemma 4 and (12), whose evaluation point z_h lies in $\boldsymbol{D}'$, give $|\boldsymbol{V}_{\theta^*} - S_h[\boldsymbol{V}_{\theta^*}]|(z) \leq |\boldsymbol{V}_{\theta^*} - \boldsymbol{V}|(z) + |\boldsymbol{V}_{\theta^*} - \boldsymbol{V}|(z_h) \leq 2\varepsilon$.

Step 4: perturbed nodes. The additional error is $|\sum_j (\tilde{c}_j - c_j)\tilde{\varphi}_j| \leq 2^m \cdot 2\eta$, and Step 3 applies unchanged. □

Remark 9 (Scope of Theorem 8)**.** After the geometric and regularity parameters $(\boldsymbol{B}, \boldsymbol{D}', r, L)$ are fixed, the only coefficients carrying value data are the nodal TTC values c_j, obtained offline by direct simulation or by the sweeps of Proposition 6; accuracy, size and depth are known before any computation, and no loss is minimized. The bound (16) is an a priori complexity estimate with an unspecified C_m, exponential in m as expected for Lipschitz functions [25]. The approximation is local to a compact subset of the transversal collision domain, separated from grazing, from the collision set and from the terminal horizon, where L or $1/r$ blow up. The theorem does not assert that $\boldsymbol{V}_{\theta^*}$ satisfies the stopped-step equation exactly, only the residual bound (18), which feeds Theorem 5 on sets $\boldsymbol{A} \subset \boldsymbol{D}$ that contain their pre-contact trajectories.

V. Numerical Illustration

We consider the OVM leader–follower pair (5) with $\alpha = 2$, $\boldsymbol{T}_h = 1$ and the piecewise constant leader input $\mathsf{a_L} = 0.5, -1.2, 0.8, -0.4, 0$ on the fifths of $[0,1]$, which is bounded and discontinuous in t as Assumption 1 allows. Samples $(s, \Xi, \Lambda, \zeta_{\mathsf{L}})$ are drawn uniformly from $[0,1) \times (0,2] \times [-3,1] \times [0,2]$, $5 \times 10^5 / 5 \times 10^4 / 10^5$ for training, validation and testing. Reference values of $\boldsymbol{V}$, impact points and the stopped flow are computed by classical RK4 with step 10^{-3} aligned with the breakpoints of $\mathsf{a_L}$ and bisection of the event $\Xi = 0$; 11.2% of the samples collide within the horizon. The compact set $\boldsymbol{D}$ is defined by $\boldsymbol{V} \geq 0.1$, a verified contact with $\Lambda(\boldsymbol{V}; z) \leq -0.8$ and $\boldsymbol{V} \leq \boldsymbol{T}_h - s - 0.2$ (3.3% of the samples); the transversality margin is what keeps the

TABLE I

TRAINED NETWORKS VERSUS THE CONSTRUCTED NETWORK (15), $h = 0.05$. SIZE: PARAMETERS, OR NODES (RELU UNITS) FOR (15); L^∞: MAXIMUM ERROR ON THE WHOLE TEST SET; $L^\infty(\boldsymbol{D})$, RMSE($\boldsymbol{D}$): ON THE TRANSVERSAL COMPACT SET; δ_h: MAXIMUM STOPPED-STEP RESIDUAL ALONG THE TRAJECTORIES FROM $\boldsymbol{D}$; BOUND: $\max_{\boldsymbol{D}} \lceil \boldsymbol{V}/h \rceil \delta_h + b$ OF THEOREM 5; ε: A PRIORI BOUND (17).

	size	L^∞	$L^\infty(\boldsymbol{D})$	RMSE($\boldsymbol{D}$)	δ_h	bound / ε
$\boldsymbol{V}_\theta$	323,201	0.93	0.0078	0.0019	0.0153	0.20
$\boldsymbol{V}_\theta^S$	323,201	0.90	0.0062	0.0015	0.0112	0.15
(15), $h_0 = 0.02$	469k ($5.9 \cdot 10^7$)	–	0.0036	0.0003	0.0038	0.120
(15), $h_0 = 0.005$	583k ($8.4 \cdot 10^7$)	–	0.0005	0.0001	0.0006	0.026

r-neighborhood of Section IV, $r = 0.02$, inside $\mathbb{D}$, since a contact with $|\Lambda| = 0.8$ under the follower's deceleration is only a few 10^{-2} in gap away from a miss (verified on 1.5×10^5 perturbed points). Errors are reported on $\boldsymbol{D}$ and on the whole test set, which includes grazing and horizon-censored states where $\boldsymbol{V}$ jumps to $\boldsymbol{T}_h - s$ and no continuous approximation has small sup-norm error.

Trained networks. $\boldsymbol{V}_\theta$ is a ReLU network with three hidden layers of width 400 (323,201 parameters) trained on the mean-squared error to the reference values, with the 55,678 impact points of the training trajectories added with target 0; $\boldsymbol{V}_\theta^S$ has the same architecture and adds the mean-squared stopped-step residual (11) with $h = 0.05$, weight 1, evaluated on the cached landing points. Both use Adam, 150 epochs, batches of 4096, learning rate 10^{-3} with cosine decay.

Constructed network. The network (15) is built with $\boldsymbol{D}' = \boldsymbol{D} \cup \boldsymbol{D}^h$, $h = 0.05$, on the box $\boldsymbol{B}$ circumscribing the r-neighborhood of $\boldsymbol{D}'$ ($\ell_{\boldsymbol{B}} = 2.15$), with meshes $h_0 \in \{0.02, 0.005\} \le r$, nodal values from direct simulation and no training. Only the $|\mathcal{J}|$ nodes whose support meets the sampled $\boldsymbol{D}'$ are kept; the empirical Lipschitz constant of $\boldsymbol{V}$ on these nodes, $L = 1.3$–1.5, gives the a priori accuracy $\varepsilon = 2L\sqrt{m}\, h_0$ and the tolerance $\varrho = \varepsilon/(2^{m+1}\boldsymbol{T}_h)$ of the proof. The unit count in Table I is that of a literal realization ($3m$ units per node for the hats, $3(6k+2)$ for the Yarotsky products with $k = \lceil \frac{1}{2}\log_2(1/\varrho) \rceil + 1$ sawtooth layers); the evaluation uses the exact interpolant, within $\varepsilon/2$ of it.

Findings. *Certificate.* For every $h \in \{0.025, 0.05, 0.1, 0.2\}$ and both trained networks, the pointwise bound (14) holds at all 3,314 test points of $\boldsymbol{D}$, with $b \le 0.0034$; the ratio of observed error to bound is at most 0.10–0.44, and the bound tightens as h grows (Fig. 2(b)), because $\delta_h \approx 0.01$ is set by the accuracy of the network rather than by h while $\lceil \boldsymbol{V}/h \rceil$ shrinks. The residual term in training lowers δ_h by 27% and $L^\infty(\boldsymbol{D})$ by 21% at equal size. *Sweeps.* On a grid of 6.6×10^6 nodes (mesh 0.05 over the sampling box padded by 0.5), S_h is applied with multilinear interpolation at the landing points. From $W_0 = 0$, after k sweeps the error on the 91,249 nodes of $\boldsymbol{D}$ with $\boldsymbol{V} \le kh$ has RMSE ≤ 0.0046 and maximum ≤ 0.048 for all $k \le 20$, the interpolation error of Proposition 6's remark, while on $\boldsymbol{V} > kh$ it equals $\boldsymbol{V} - kh$ up to interpolation, the front visible in Fig. 2(a). From the constant $W_0 \equiv 0.5 \notin \mathscr{B}_0$ the same holds one sweep later, as stated in Proposition 6: at $k = 10$ the slab $9h < \boldsymbol{V} \le 10h$ still carries an RMSE of 0.24 while $\boldsymbol{V} \le 9h$ has 0.020. *Subsolutions.* For $W = \lambda \boldsymbol{V}$ on $\boldsymbol{D}$ the optional-duration residual of Proposition 7 is 0 for $\lambda \in \{0.5, 0.9, 1\}$ and 0.005 for $\lambda = 1.1$, while the stopped-step residual is $(1-\lambda)h$ exactly and the error $|1-\lambda| \max \boldsymbol{V}$ reaches 0.31 for $\lambda = 0.5$. *Construction.* The constructed network meets (17) on $\boldsymbol{D}'$ at both meshes (0.0041 and 0.0008 against $\varepsilon = 0.120$ and 0.026), the factor 30 being the gap between a Lipschitz bound and the second-order accuracy of linear interpolation; its residual on $\boldsymbol{D}$ is below 2ε as in (18); and perturbing all nodal values by $\eta = 0.01$ changes it by at most $0.0094 \le 2^{m+1}\eta$. At equal accuracy on $\boldsymbol{D}$ it is two orders of magnitude larger in units than the trained networks, the price of an optimization-free guarantee, and it is defined only near $\boldsymbol{D}'$ (Fig. 1(c)). All networks have sup-norm errors near 1 over the whole test set, concentrated at the grazing boundary of Fig. 1(a), where $\boldsymbol{V}$ is discontinuous and Theorem 8 makes no claim.

VI. CONCLUSION

For a feedback law defined through the first contact time of a prediction model, we proved local Lipschitz regularity of the value function on the transversal collision domain for Carathéodory flows with bounded inputs, and characterized it by a stopped-step Bellman equation that involves no derivatives, has a unique solution, and comes with a two-sided a priori certificate in terms of the residual along trajectories; exact recovery holds after finitely many sweeps, and an explicit ReLU realization of known size exists. Future work includes a complete numerical error budget for the certificate, learning with the stopped-step residual as the sole objective, and the transfer of approximation guarantees to the closed loop (1).

APPENDIX

APPENDIX: UNIFORM BOUNDEDNESS UP TO CONTACT

Lemma 10. *Let (A1)–(A2) hold and let $\boldsymbol{C}_0 \subset P$ be compact. There is a compact $\boldsymbol{C}_1 \subset P$ such that every maximal solution in P from a datum in $[0, \boldsymbol{T}_h] \times \boldsymbol{C}_0$ takes its values in $\boldsymbol{C}_1$; in particular $\Phi(\kappa; s, \boldsymbol{y}) \in \boldsymbol{C}_1$ for all $(s, \boldsymbol{y}) \in [0, \boldsymbol{T}_h] \times \boldsymbol{C}_0$, $0 \le \kappa \le \varsigma(s, \boldsymbol{y})$.*

Proof. Two consequences of (A1). *(F1)* If x, x' are solutions in P on $[a, b]$ with values in a compact $\boldsymbol{K} \subset P$, then $|x(t) - x'(t)| \le e^{L_{\boldsymbol{K}}(t-a)}|x(a) - x'(a)|$ on $[a, b]$ and $|x(t) - x(t')| \le M_{\boldsymbol{K}}|t - t'|$. *(F2)* If x_n are solutions in P on $[b_n, b_n + \theta]$ with $b_n \to b$, values in a compact $\boldsymbol{K} \subset P$, and $x_n(b_n) \to \bar{\boldsymbol{y}}$, then a subsequence converges uniformly on $[b, b+\theta/2]$, after extension by the constant $x_n(b_n)$ on $[b, b_n]$ when $b_n > b$, to a solution in P on $[b, b + \theta/2]$ from $(b, \bar{\boldsymbol{y}})$: the extended functions are $M_{\boldsymbol{K}}$-Lipschitz and bounded, so Arzelà–Ascoli gives a uniformly convergent subsequence with limit $\hat{x}$, with values in the closed set $\boldsymbol{K}$; for $t \in (b, b+\theta/2]$ and n large, $x_n(t) = x_n(b_n) + \int_{b_n}^t \mathsf{G}(r, x_n(r))\, dr$, and since $\mathsf{G}(r, \cdot)$ is

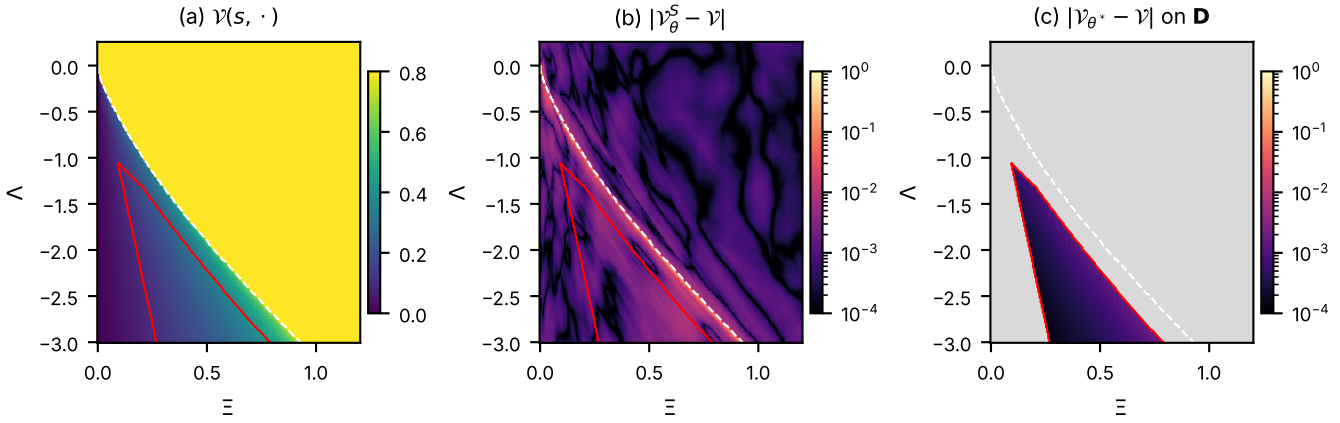


Fig. 1. Slice $s = 0.2$, $\zeta_{\mathrm{L}} = 1$ of the (Ξ, Λ) plane: (a) reference TTC, with the grazing boundary (white, dashed) beyond which $\boldsymbol{V} = \boldsymbol{T}_h - s$, and the boundary of $\boldsymbol{D}$ (red); (b) absolute error of $\boldsymbol{V}_\theta^S$; (c) absolute error on $\boldsymbol{D}$ of the network (15) built for this slice with $h_0 = 0.02$ (18,653 nodes), undefined elsewhere.

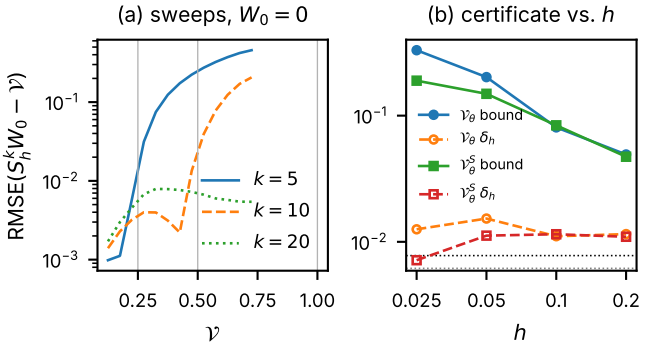


Fig. 2. (a) RMSE of $S_h^k[0] - \boldsymbol{V}$ on the nodes of $\boldsymbol{D}$ by level of $\boldsymbol{V}$, mesh 0.05, $h = 0.05$; vertical lines at $\boldsymbol{V} = kh$. (b) Certificate bound and residual δ_h of the trained networks against h; dotted: observed $L^\infty(\boldsymbol{D})$ errors.

Lipschitz on $\boldsymbol{K}$ for a.e. r and bounded by $M_{\boldsymbol{K}}$, dominated convergence gives $\hat{x}(t) = \bar{\boldsymbol{y}} + \int_b^t \mathsf{G}(r, \hat{x}(r))\, dr$.

Suppose the lemma fails. Let $P_1 \subset P_2 \subset \cdots$ be compact subsets of P with $\bigcup_k P_k = P$ and P_k contained in the interior of P_{k+1} relative to P (such an exhaustion exists since P is closed in the open set U). Then for every k some maximal solution x_k in P from $(s_k, \boldsymbol{y}_k) \in [0, \boldsymbol{T}_h] \times \boldsymbol{C}_0$, on $[s_k, s_k + \sigma_k]$, satisfies $x_k(t_k) \notin P_k$ for some t_k. Passing to a subsequence, $(s_k, \boldsymbol{y}_k) \to (s^*, \boldsymbol{y}^*)$. Let x^* be the maximal solution in P from $(s^*, \boldsymbol{y}^*)$, on $[s^*, e^*]$; by (A2) its graph is compact. Fix $\eta \in (0, \operatorname{dist}(x^*([s^*, e^*]), \partial U))$, with $\operatorname{dist}(\cdot, \varnothing) = \infty$, and set $\boldsymbol{K} := \{\boldsymbol{y} \in P : \operatorname{dist}(\boldsymbol{y}, x^*([s^*, e^*])) \le \eta\}$, a compact subset of P since the closed η-neighborhood of the graph is a compact subset of U and P is closed relative to U; let $M := M_{\boldsymbol{K}}$, $L := L_{\boldsymbol{K}}$. We show that x_k takes values in $\boldsymbol{K}$ for all large k.

Let $\delta_k := |\boldsymbol{y}_k - \boldsymbol{y}^*| + M|s_k - s^*| \to 0$ and $T_k := \sup\{t \in [s_k, s_k + \sigma_k] : x_k([s_k, t]) \subset \boldsymbol{K}\}$; for k large $\boldsymbol{y}_k \in \boldsymbol{K}$, and $x_k([s_k, T_k]) \subset \boldsymbol{K}$ by continuity. Assume $T_k < s_k + \sigma_k$ for infinitely many k; we show that $x_k(T_k)$ then lies within $\eta/2$ of $x^*([s^*, e^*])$, so that x_k, being M-Lipschitz while in $\boldsymbol{K}$, remains in $\boldsymbol{K}$ on $[T_k, \min\{T_k + \eta/(2M), s_k + \sigma_k\}]$, contradicting the definition of T_k.

Step 1: the overlap. Let $a_k := \max\{s_k, s^*\}$ and suppose $a_k \le \min\{T_k, e^*\}$. On $[\min\{s_k, s^*\}, a_k]$ only one solution moves, inside $\boldsymbol{K}$, so by (F1) $|x_k(a_k) - x^*(a_k)| \le \delta_k$, and on $[a_k, \min\{T_k, e^*\}]$ both lie in $\boldsymbol{K}$, so

$$|x_k(t) - x^*(t)| \le e^{L\boldsymbol{T}_h}\delta_k, \qquad t \in [a_k, \min\{T_k, e^*\}]. \quad (19)$$

If $T_k \le e^*$ and $a_k \le T_k$, (19) at $t = T_k$ gives the announced contradiction for k large. If $T_k < a_k$ (only if $s_k < s^*$), then $T_k - s_k < s^* - s_k$ and $|x_k(T_k) - \boldsymbol{y}^*| \le \delta_k$, again a contradiction. Hence $T_k > e^*$ for k large.

Step 2: beyond e^*. Since $T_k \le \boldsymbol{T}_h$, $e^* < \boldsymbol{T}_h$, so x^* ends at e^* and admits no continuation in P. Let $b_k := \max\{s_k, e^*\}$ and $\theta_k := T_k - b_k > 0$. By (19) at $t = e^*$ when $s_k \le e^*$, and by $x_k(b_k) = \boldsymbol{y}_k$ otherwise (then $e^* = s^*$), $x_k(b_k) \to x^*(e^*)$. If $\theta_k \ge \theta > 0$ along a subsequence, (F2) applied to the restrictions of x_k to $[b_k, b_k + \theta]$ yields a solution in P on $[e^*, e^* + \theta/2]$ from $(e^*, x^*(e^*))$, whose concatenation with x^* contradicts maximality. Hence $\theta_k \to 0$ and $|x_k(T_k) - x^*(e^*)| \le |x_k(b_k) - x^*(e^*)| + M\theta_k \le \eta/2$ for k large, the announced contradiction.

Therefore x_k takes values in $\boldsymbol{K}$ for all large k. Since $\boldsymbol{K}$ is a compact subset of P, it is contained in P_k for k large, contradicting $x_k(t_k) \notin P_k$. Hence there is k_0 such that every maximal solution from $[0, \boldsymbol{T}_h] \times \boldsymbol{C}_0$ takes values in P_{k_0}, and $\boldsymbol{C}_1 := P_{k_0}$ is the required set. □

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